

MUHAMMAD MUKARRAM RAZA

BS Software Engineering · Software Engineer
contact.raza.dev@gmail.com · LinkedIn · Google Scholar · Website

Professional Summary

Software Engineer focused on **Flutter mobile development**, **TypeScript/NestJS backends**, and **applied AI**. Experienced with production mobile interfaces, REST APIs, modular server architecture, computer vision research, and cloud deployment. Comfortable moving between user-facing product work and the backend systems that support it.

Technical Skills

	Proficient	Familiar
Languages	Dart, Python, C, C++, JavaScript, SQL	TypeScript, Java, C#, Rust, Go, PHP, RISC-V
Web & Mobile	Flutter, REST APIs, React, Node.js, NestJS	Next.js, Tailwind CSS, Svelte
AI & ML	CNNs, RNNs, transfer learning, computer vision	TensorFlow, Keras, LangChain, OpenCV, Scikit-learn, Chroma, Pinecone
Systems & Arch.	OOP, DSA, multithreading, design patterns	async programming, gRPC, OS internals (xv6), Win32 API
Cloud & DevOps	AWS (Lambda, API Gateway, DynamoDB, S3), Docker	Kubernetes, GCP, Jenkins, Redis, Cloudflare
Databases	PostgreSQL, MySQL, MongoDB, SQLite	Firebase, Supabase, Redshift
Tooling	Git, Bash, VS Code, JetBrains, Ubuntu/WSL, LaTeX	CI/CD, Android Studio, Xcode, Postman, QEMU

Experience

Flutter Developer Intern | *Granjur Technologies (Lahore, Pakistan)* (3months) Jun 2025 – Aug 2025

- Contributed to production-grade mobile applications within a professional technology firm.
- Architected and scaled a 50 KSLOC codebase using Flutter, implementing custom animations and complex stateful UI.

Projects

Serverless URL Shortener | *Mobile & Cloud Development* Flutter, Dart, AWS

- Built a cross-platform Flutter client with authentication, MFA, dashboards, URL analytics, optimistic UI, and a serverless AWS architecture.

Multilingual Reading and Search App | *Cross-Platform Application* Flutter, Dart, Rust, Cloudflare

- Developed a multilingual cross-platform reading application with audio, bookmarks, localization, and Rust-backed vector search.

Tuberculosis Classification Pipeline | *Medical AI & Computer Vision* Python, TensorFlow, Keras

- Built a reproducible VGG16 transfer-learning pipeline for three-class chest X-ray classification, reporting 98.53% accuracy and 97.54% MCC; published at IEEE ICIC 2025.

TypeScript Backend and Microservices | *Backend Engineering* TypeScript, NestJS, Node.js

- Built modular server components with decorators, controllers, route handlers, middleware, and service-oriented structure; applied the same backend concepts while learning NestJS.

Traceable PDF Agents | *Applied AI & Backend* Python, LangChain, Langfuse, Redis

- Prototyped retrieval-augmented PDF agents with Flask, worker processes, Redis, vector stores, and tracing-oriented integrations.

Hotel Management Systems | *Full-Stack Web Development* React, Node.js, PostgreSQL, AWS

- Shipped web application implementations with React, Node.js, PostgreSQL, and PHP, covering structured data management and user-facing workflows.

Network Traffic Dissector | *Security & Network Tooling* Go, PCAP, TCP/IP

- Built a passive layer-by-layer traffic analyzer with protocol decoding, HTTP inspection, PCAP replay, JSONL output, and golden tests.

Publications

An Experimental Study for Tuberculosis Classification Using Efficient VGG16 Transfer Learning Pipeline

IEEE - 2025

2025 6th International Conference on Innovative Computing (ICIC) · DOI: 10.1109/ICIC68258.2025.11413008

Education

Bachelor of Science – Software Engineering

2023 – Present

Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Organization & Assembly, Software Engineering, Software Design Architecture, Software Design & Construction, Software Re-engineering, Software Project Management, Analysis of Algorithms, Theory of Automata, Artificial Intelligence, Machine Learning, Cloud Computing, Information Security, Human-Computer Interaction, Linear Algebra, Multivariable Calculus, Discrete Mathematics, Probability & Statistics.